

Question number	Answer	Additional guidance	Mark
3(a)(i)	• D is the correct answer A is not the correct answer as baroreceptors detect pressure changes B is not the correct answer as chemoreceptors detect changes in chemical stimuli C is not the correct answer as osmoreceptors detect changes in osmolarity stimuli		(1)

Question number	Answer	Additional guidance	Mark
3(a)(ii)	A description that includes three of the following points: - receptors synapse with sensory neurones (1) - impulse transmitted via sensory neurone towards the {CNS / spinal cord} (1) - which passes through dorsal root ganglion (1) - where sensory neurone synapses with {relay / motor} neurone (1) (synapse / relay neurone) located in the grey matter (1)	accept impulse transmitted along sensory neurone accept impulse / action potential transmitted via sensory neurone to relay neurone	(3)

Question number	Answer	Additional guidance	Mark
3(b)(i)	A calculation showing the following steps: - correct difference from the graph (1) - correct percentage difference (1)	$20 - 12 = (-) 8$ $(8 \div 16) \times 100 = (-) 50\%$ OR $0.5 \times 100 = (-) 50\%$ correct answer gains both marks	(2)

Question number	Answer	Additional guidance	Mark
3(b)(ii)	A description that includes three of the following points - all the groups {recover / start changing direction} after 0.5 mins (1) - recovery increases as time increases (1) - steeper increase up to 10 mins / slower increase after 10 mins (1) - the shorter the habituation interval the {faster the recovery time / higher number of worms changing direction} (1) - all three groups have worms that have not {recovered / started changing direction} (after 30 mins)ii(1)	accept the number of roundworms changing direction increases as time increases {gradient / graph} steeper up to 10 minutes accept converse accept group {1 / 60 taps at 2s intervals} had the highest number changing direction / group {3 / 60 taps at 60s intervals} had the lowest number changing direction	(3)

